

DAY 1 · SATURDAY, 11 JULY 2026



WELCOME TO

Agentic Software Design & Architecture

BOOTCAMP

Let's build engineers who design.

THE NEXT 60 MINUTES

This First Hour

1	Welcome & why we're here	10 min
2	Meet Backendguru and your four trainers	10 min
3	The journey & the full schedule	15 min
4	How it works & ground rules	10 min
5	Your turn — introduce yourselves	15 min

...then we dive straight into the first lesson.

Why This Program Exists

Reach developers at the start of their careers — and give them a broad, solid foundation in software design and architecture.

<https://www.backendguru.com/products/software-design-and-architecture-bootcamp>



Accessible

A first-of-its-kind bootcamp in the country, priced so students and juniors can actually join.



Foundational

Shallow but wide: the concepts every engineer keeps using — cohesion, coupling, design, architecture.



AI-native

You learn to design and build with AI as a partner — while you stay the decision-maker.

Who's Here · What You'll Gain

Who's in the room

- Students & new graduates
- Junior backend developers (1–2 years)
- Experienced developers who never had formal design training

By the end, you'll be able to

- Reason about cohesion, coupling & change
- Apply Clean Code, SOLID & core design patterns
- Design REST / gRPC APIs and microservices
- Weigh architectural quality attributes & trade-offs
- Lead design & coding with AI — as the decision-maker

SECTION 01

Meet Backendguru and Your Trainers

Four practitioners — one journey from clean code to cloud architecture.

Can Yasun

Backendguru

- Who is Backendguru?
- <https://www.backendguru.com/>

<https://www.linkedin.com/in/canyasun/>

Akın Kaldıroğlu

TEACHES

- Introduction to SW Design & Architecture
- Fundamentals of SW Design · Clean Code · SOLID
- Design Patterns · Secure Coding · Developer Testing
- Agentic Design & Coding · SW Design Project with AI
- Fundamentals of SW Architecture
 - Layered · Hexagonal · Onion · Clean Architecture

<https://github.com/kaldiroglu>

<https://www.linkedin.com/in/akinkaldiroglu/>

My Bootcamp Repos

All the course code lives on GitHub — clone it, run the tests, break it, refactor it. You can send your PRs too.

Companion code · Java *smell → fix, with tests*

▶ github.com/kaldirglu/Agentic-Software-Design-and-Architecture-Bootcamp-JAVA

Companion code · C# / .NET *smell → fix, with tests*

▶ github.com/kaldirglu/Agentic-Software-Design-and-Architecture-Bootcamp-DOTNET

Companion code · Python *smell → fix, with tests*

▶ github.com/kaldirglu/Agentic-Software-Design-and-Architecture-Bootcamp-PYTHON

Refactoring exercises *3 projects (C#, Java, Python) for the hands-on exercises*

▶ github.com/kaldirglu/refactoring-course-student-materials

Emre Akış

TEACHES

- Introduction to Distributed Architecture: SOA & Microservices
- API Design & Microservices
- From monoliths to distributed systems — risks & benefits

<https://github.com/emreakis>

<https://www.linkedin.com/in/emreakis/>

Alper Hankendi

TEACHES

- **Data Architecture: Event Sourcing**
- **System Design Project with AI · Spec-driven development**
- **System Design Interview — how to prepare & perform**

<https://github.com/alperhankendi/>

<https://www.linkedin.com/in/alperhankendi/>

Fatih Çakıroğlu

TEACHES

- Security Architectures — secure app development & web security
- Cloud Architectures — AWS · Google Cloud · Microsoft Azure

<https://github.com/Fcakiroglu16>

<https://www.linkedin.com/in/fcakiroglu16/>

The Journey — Two Phases

PHASE 1

Software Design

Small-scale · foundations · ~20 h

- Design fundamentals · cohesion & coupling
- Clean Code · SOLID · Design Patterns
- Secure Coding · Developer Testing
- Agentic coding + a design project with AI

PHASE 2

Software Architecture

Large-scale · systems · ~24 h

- Architecture fundamentals & models (4+1, C4)
- Layered · Hexagonal · SOA · Microservices
- API design · Data & Event Sourcing
- Security · Cloud ·
- System design project & interview

~44 hours · 8 weeks · July → September 2026

Schedule — Phase 1: Software Design

DATE	TOPIC	TRAINER
Sat 11 Jul	Introduction to SW Design & Architecture	Akın
Tue 14 Jul	Fundamentals of SW Design	Akın
Sat 18 Jul	Clean Code & SOLID	Akın
Tue 21 Jul	SOLID	Akın
Sat 25 Jul	Design Patterns & Secure Coding	Akın
Tue 28 Jul	Developer Testing	Akın
Sat 1 Aug	Agentic Design & Coding + SW Design Project with AI	Akın

Today: after this welcome hour, we begin 'Introduction to SW Design & Architecture'.

Schedule — Phase 2: Software Architecture

DATE	TOPIC	TRAINER
Tue 4 Aug	Fundamentals of SW Architecture	Akın
Sat 8 Aug	Layered & Hexagonal / Onion / Clean Architecture	Akın
Tue 11 Aug	Intro to Distributed Arch: SOA & Microservices	Emre
Sat 15 Aug	API Design & Microservices	Emre
Tue 18 Aug	Data Architecture: Event Sourcing	Alper
Sat 22 Aug	Security & Cloud Architectures	Fatih
Tue 25 Aug	System Design Project with AI	Alper
Sat 29 Aug	System Design Interview	Alper
Tue 8 Sep	Project Presentations	Everyone

How It Works

Cadence

Saturdays 10:00–14:00 (4 h)

Tuesday evenings 20:00–22:00 (2 h)

Duration

~44 hours over 8 weeks (Jul–Aug)

Presentations on 8 Sep

Languages

Examples in C#, Java & Python

AI-first

Work with any LLM model — always you stay the decision-maker

Capstone

A real business problem, in small groups (4–5)

10–15 min final presentations

Everything on GitHub

Code, exercises and projects are shared and reviewed on GitHub

Your Certificates

You leave the bootcamp with proof of your work — and finishing your project earns you more.

EVERY PARTICIPANT

Certificate of Participation

Every participant in the bootcamp receives a Certificate of Participation.

PROJECT + PRESENTATION

Certificate of Completion

Complete your capstone project and present it to earn a second certificate recognising your completed project — in addition to your participation certificate.

Ground Rules & Logistics

01 Show up

Be present and on time: Saturdays 10:00, Tuesdays 20:00

02 Speak up

Cameras and mics welcome — this is a conversation, not a broadcast.

03 Ask anything anytime

No such thing as a dumb question. Do not ask a permission to ask a question!

04 Practice

Work between sessions — design is a muscle you build by using it.

05 Recordings

Sessions will be published to our LMS channel in the same day.

06 Stay connected

Community channel: WhatsApp

SECTION 02

Your Turn — Introduce Yourself 🖐️

Tell us who you are — keep it short so everyone gets a turn.

■ -Your Turn

Look! Who is here?

Why are you here?

- Who wants to introduce herself and tell us why she is here?
- What are your expectations from this bootcamp?
- What makes this bootcamp perfect for you?

Let's get connected: Please share your LinkedIn profile in Whatsapp channel

What's Next?

LESSON 1 · TODAY

Introduction to Software Design & Architecture

with Akin Kaldıroğlu

What design is, what architecture is — and why they are not the same thing.

Grab water, stretch — back in 15 minutes.

Let's Begin.

Eight weeks. From clean code to cloud. Let's design.